Using the ARMADA Algorithm to Analyze Physiological Marker Patterns in Presumed Emotional States

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Abstract

Affective computing has emerged as a pivotal field in bridging the gap between human emotional intelligence and machine responsiveness, focusing on the computational modeling and automatic recognition of human emotions. This study aims to investigate the feasibility of detecting sequential patterns of physiological transitions during presumed emotional episodes. To achieve this, the ARMADA algorithm is utilized to analyze datasets characterized by continuous annotations and synchronized cardiovascular and electrodermal features. Our findings include [to be completed based on experimental results]. [Final concluding sentence regarding the impact of the findings to be added here].

Keywords: Physiological Rule Mining, Emotion Recognition, ARMADA Algorithm, Affective Computing, Cross-Dataset Analysis

1. Introduction

Emotion recognition plays a crucial role in Affective Computing, offering the potential for continuous assessment of human emotional states through physiological signals [17]. While numerous scientific works have successfully extracted temporal and sequential rules to model emotions, these efforts have predominantly focused on text data, such as social media posts and blogs [20, 22, 4], or video data [15].

Surprisingly, the application of sequential rule mining to physiological signals remains virtually unexplored. Even though human physiological responses naturally unfold over time, very few studies actually use sequential rules to analyze them. Instead, modern research frequently relies on machine learning and deep learning architectures to process physiological time-series data, including Support Vector Machines (SVMs), Convolutional Neural Networks (CNNs), and Recurrent Neural Networks such as LSTMs. Although these models often achieve strong classification performance, their predictions are typically difficult to interpret in mechanistic terms, making it challenging to explicitly explain how temporally evolving physiological changes relate to the emergence of a particular emotional state [1, 19].

2. Motivations and Research Questions

To address this gap, this paper introduces a sequential rule mining approach to physiological data, shifting the focus from opaque "black-box" predictions to interpretable temporal rules. Specifically, this study aims to answer the following research questions:

1. Are there universal temporal patterns in which changes in physiological signals precede changes in emotional states (arousal, valence)?

2. Do temporal rules work bidirectionally (physiology → emotions and emotions → physi-

ology), or does one direction dominate?

3. Do the extracted temporal patterns differ between genders?
4. Does age affect the discovered physiological rules?
5. Which temporal rules extracted by the ARMADA algorithm remain stable regardless of the experimental context?
6. Do multimodal rules (combining EDA, HRV, and HR simultaneously) exhibit higher confidence than unimodal rules?

3. Related Work

3.1. Emotion Recognition and Physiological Signals

The field of Affective Computing has long recognized the potential of physiological signals for emotion assessment. Unlike facial expressions or voice, physiological responses such as Electrodermal Activity (EDA), Photoplethysmography (PPG), and Heart Rate Variability (HRV) are largely controlled by the autonomic nervous system and reflect genuine emotional states without conscious distortion [19]. In recent years, an extensive body of literature has adopted machine learning and deep learning techniques to process these complex time-series signals [1]. Studies frequently rely on sophisticated architectures capable of high-accuracy classification. However, while these predictive models demonstrate impressive performance, they predominantly function as black-box mechanisms, rarely offering transparent insights into how temporal interactions among distinct physiological events shape human emotions over time.

3.2. Sequential Rule Mining in Affective Computing

While sequential pattern mining has been successfully applied to discover temporal relationships within data, its utilization in emotion recognition has been mostly restricted to other modalities. A significant amount of research has focused on extracting sequential emotion-aware rules from text data, particularly social media content and micro-blogs [20, 22]. For instance, tracing the temporal emotional roadmap of users through sequences of online posts has proven valuable in depressive tendency detection [4]. Similarly, sequential pattern mining has been utilized to analyze multimodal combinations of verbal and facial clues in video data, unearthing temporal patterns during interpersonal interactions [15]. Despite these advancements across text and video modalities, physiological data remains notably excluded from sequential rule-mining approaches, leaving an unaddressed gap regarding the explicit, universally interpretable temporal rules defining human physiological reactions.

4. Methodology

4.1. Datasets

Although a broader range of affective datasets exists [12], in this work we focused on those that combine multimodal physiological recordings with relatively continuous or densely sampled emotion annotations and for which data access could be obtained.

The analysis was conducted on five multimodal emotion datasets: EMBOA, K-emoCon, CASE, CEAP-360VR, and EmoWorker [8, 13, 17, 24, 10]. These datasets differ in both acquisition setting and annotation protocol, covering controlled laboratory recordings as well as conversational and immersive VR scenarios. They also include both self-reported annotations and externally assigned labels, which makes them suitable for evaluating rule mining under heterogeneous emotion-labeling schemes.

More specifically, EMBOA was collected in a human-robot interaction setting and combines Empatica E4 peripheral signals with externally assigned behavioral annotations [8], whereas K-emoCon captures dyadic conversations and provides self, partner, and external affective ratings alongside physiological recordings [13]. CASE represents a controlled laboratory dataset with high-resolution physiological signals and continuous self-assessment gathered using a joystick interface [17], while CEAP-360VR extends this paradigm to immersive 360° audiovisual stimulation in virtual reality [24]. EmoWorker, in turn, provides multimodal physiological recordings acquired in a workplace-related context together with continuous affect annotations [10]. The annotation schemes used in the evaluated datasets are summarized in Table 1.

Table 1. Annotation methodologies and frequencies across evaluated datasets.

Dataset	Annotation Type	Annotation Freq.
EMBOA	External (BORIS)	Variable
K-emoCon	Self, Partner, External	0.2 Hz (every 5s)
CASE	Self (Continuous Joystick)	20 Hz
CEAP-360VR	Self (Continuous + SAM Questionnaire)	10 Hz
EmoWorker	Self (Continuous traces)	5-10 Hz

4.2. Modalities

The datasets further vary in the physiological modalities and sampling rates they provide. While several of them share common peripheral signals, such as Electrodermal Activity (EDA) and Blood Volume Pulse (BVP), others additionally include modalities such as ECG, EEG, respiration, eye tracking, or motion-related signals [8, 13, 17, 24, 10]. The physiological modalities considered in our experiments are listed in Table 2.

Table 2. Primary physiological modalities and sampling frequencies across evaluated datasets.

Dataset	Physiological Frequencies
EMBOA	EDA (4Hz), BVP (64Hz), ACC (32Hz), SKT (4Hz)
K-emoCon	EDA (4Hz), BVP (64Hz), ACC (32Hz), SKT (4Hz), EEG, ECG
CASE	EDA, BVP, ECG, RSP, EMG, SKT (All 1000Hz)
CEAP-360VR	EDA (4Hz), BVP (64Hz), ACC (64Hz), SKT (4Hz), Eye tracking
EmoWorker	EDA (4Hz), BVP (64Hz), ACC (32Hz), SKT (4Hz), ECG (130Hz), EEG (256Hz)

To facilitate a unified comparative analysis and enable the ARMADA-based rule mining process, data from all disparate sources was resampled, synchronized, and aligned into standardized 5-second multi-resolution time windows.

The common modalities used in the cross-dataset analysis were EDA, BVP, and skin temperature (SKT). These signals were selected because they are available in most of the analyzed datasets and provide some information about autonomic aspects of emotional responses.

Electrodermal activity (EDA) reflects changes in skin conductance driven mainly by sympathetic nervous system activation [7]. For this reason, EDA is one of the physiological modalities most consistently associated with arousal [7, 18].

Blood Volume Pulse (BVP) is a pulse-related physiological signal derived from photoplethysmography [11], from which features such as heart rate (HR), inter-beat interval (IBI), and heart rate variability (HRV) can be estimated. HRV features include, for example, SDNN, RMSSD, and pNN50 [9]. In the literature, BVP-derived features are more consistently associated with arousal than with valence [14, 7, 19].

Skin temperature (SKT) is a peripheral physiological measure influenced by blood flow in the skin and autonomic nervous system activity. In emotion research, it is treated mainly as an indirect marker of autonomic arousal [16].

4.3. Sequential Association Rule Mining

Sequential rule mining is a specialized data mining framework aimed at discovering temporal associations between events in sequence databases. [3, 2] Unlike standard association rule mining (e.g., Apriori [6]), which identifies co-occurring items regardless of order, sequential rule mining extracts patterns of the form $X \to Y$, implying that the antecedent event X temporally precedes the consequent event Y. Sequential pattern mining algorithms, such as GSP (Generalized Sequential Pattern) [21] or PrefixSpan [5], focus on discovering frequent subsequences and are often used as baselines or building blocks for rule extraction.

Support quantifies the prevalence or frequency of a rule across the entire dataset. It is defined as the proportion of sequences in S where the event X is followed by the event Y:

$$Support(X \to Y) = \frac{|\{s \in S \mid X \text{ precedes } Y \text{ in } s\}|}{|S|} \tag{1}$$

Confidence measures the directional reliability of the rule, defined as the conditional probability that Y will follow X, given that X has already occurred:

$$Confidence(X \to Y) = \frac{Support(X \to Y)}{Support(X)} \tag{2}$$

In the context of physiological emotion recognition, defining strict bounds for these metrics alongside explicit temporal gaps is crucial to ensure that physiological reactions are not arbitrarily linked to temporally distant emotion states.

Selection of the ARMADA Algorithm

Selecting an appropriate data mining algorithm for this study required matching the specific nature of our dataset with the algorithm's technical capabilities. Specifically, the chosen method had to support multiple data channels simultaneously, integrate both categorical variables (e.g., participant sex) and continuous numerical values (e.g., EDA signals), and, most importantly, capture strict time consequences between physiological events and emotional responses.

Although some temporal rule-mining approaches can operate without an explicit discretization step [25], this is not problematic in our setting, as ARMADA is designed for discretized interval-based events [23]. This representation is also consistent with the nature of the analyzed features, which are subject-specific and normalized before rule extraction, making their transformation into discrete states both natural and interpretable.

Moreover, ARMADA is able to represent several types of temporal relationships between interval-based events, including *before*, *meets*, *overlaps*, *is-finished-by*, *contains*, *equals*, and *starts* [23]. Thus, the relation between two events is not limited to a simple ordering such as A before B, but can also capture more specific temporal configurations between intervals. This is useful in physiological emotion recognition, where the association between a physiological response and an emotional state may depend not only on event order, but also on the exact temporal relation between them. In addition, the `MAXGAP` parameter makes it possible to constrain the maximum allowable delay between related events, focusing the rule discovery process on temporally local dependencies [23].

5. Experimental Design

The overall experimental workflow consists of five primary stages designed to extract, process, and evaluate temporal rules relating physiological signals to emotional states. The detailed

pipeline is illustrated below:

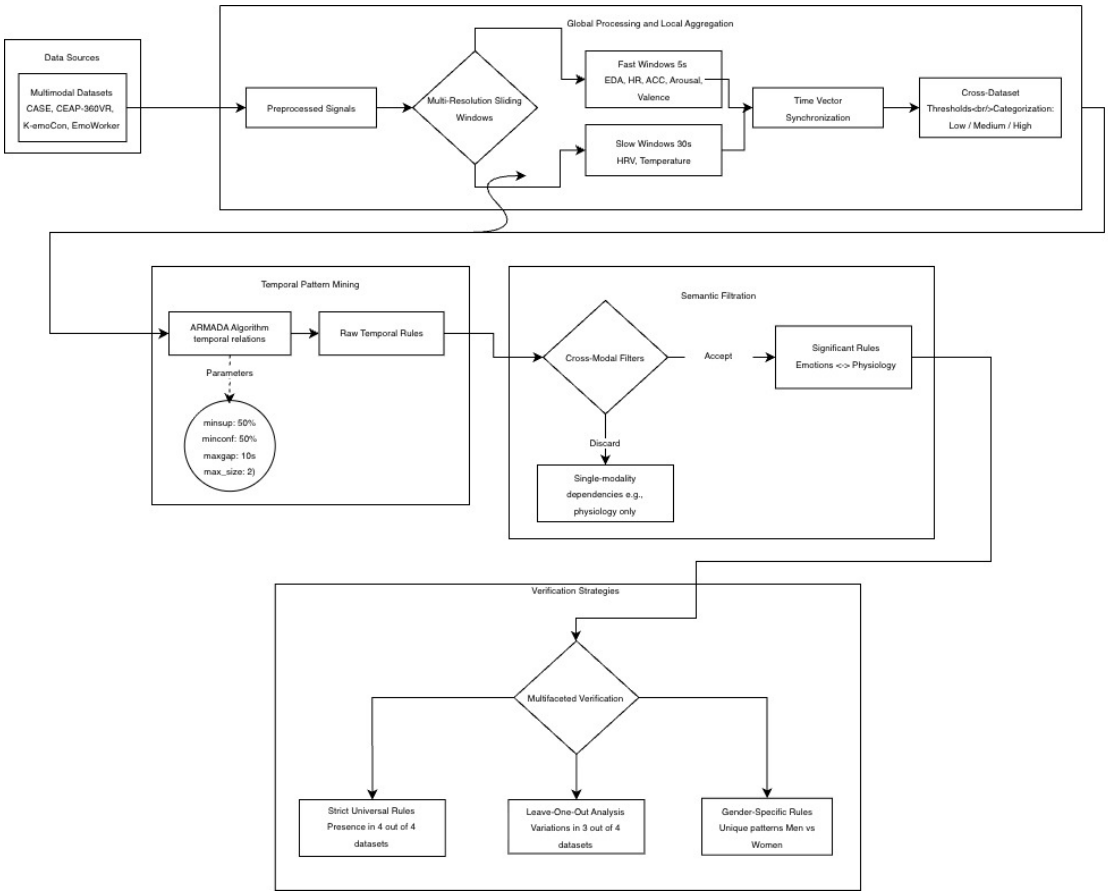

Fig. 1. The proposed experimental pipeline for extracting and verifying temporal physiological rules.

As outlined in Figure 1, the workflow involves the following steps:

1. **Data Sources:** The pipeline initially ingests multimodal physiological data from four diverse datasets: CASE, CEAP-360VR, K-emoCon, and EmoWorker.

2. **Global Processing and Local Aggregation:** The raw signals undergo global filtering, which includes peak detection and EDA decomposition, to derive clean preprocessed signals. These signals are then divided into multi-resolution sliding windows: 5-second fast windows for rapidly changing variables (EDA, HR, ACC) and 30-second slow windows for more stable measures (HRV, SKT). Finally, all time vectors are synchronized and the numerical data is discretized into cross-dataset thresholds (Low, Medium, High).

3. **Temporal Pattern Mining:** The categorized threshold data is fed into the ARMADA algorithm. Driven by specific support, confidence, and maximum gap parameters, the algorithm uncovers robust temporal relationships (such as *before*, *equals*, and *meets*).

4. **Semantic Filtration:** To ensure relevance to the study's objective, raw temporal rules are processed via cross-modal filters. Any rules that identify dependencies exclusively within a single modality (for example, a physiological change preceding solely another physiological change) are discarded, preserving only those significant rules that securely link an emotional state with corresponding physiological markers.

5. **Verification Strategies:** The remaining set of significant rules undergoes a three-pronged multifaceted verification phase: Strict Universal Rules analysis to find patterns holding across all four datasets, Leave-One-Out Analysis to spot major cross-dataset variations, and Gender-Specific checks to uncover unique behavioral variants between men and women.

6. Execution of Experiments

The ARMADA algorithm was executed independently on each of the four datasets (CASE, K-emoCon, CEAP, EmoWorker_v2). Two separate experimental runs were performed with different parameter settings to address different aspects of the research questions.

6.1. Initial Cross-Dataset Comparison

In the first run, the algorithm was configured with a minimum support threshold of 30% ($minsup = 0.3$), minimum confidence of 50% ($minconf = 0.5$), a maximum temporal gap of 5 seconds ($maxgap = 5$), and a maximum pattern size of 3. These relatively strict parameters were chosen to identify only frequently occurring and highly confident rules within each dataset individually.

6.2. Universal Rules Identification

In the second run, the parameters were relaxed to $minsup = 0.1$, $minconf = 0.1$, $maxgap =$ 20 seconds, and $max_pattern_size = 2$. The lower thresholds were intentionally set to allow weaker but potentially universal patterns to surface across all four datasets. After mining, a strict intersection was computed: only rules present in every dataset simultaneously were classified as universal.

6.3. Semantic Filtering

In both runs, the following semantic filters were applied to ensure that rules carry cross-modal meaning relevant to the research question:

- Rules containing only BVP/HRV/HR-related states were rejected.
- Rules containing only EDA-related states were rejected.
- Rules linking exclusively physiological modalities (e.g., EDA with BVP) without any emotional dimension (arousal or valence) were rejected.
- Rules involving only a single feature type were rejected.

These filters ensured that the remaining rules always connect at least one physiological modality to at least one emotional dimension.

6.4. Dataset-Level Statistics

Table 3 summarizes the number of participants and rules extracted per dataset in each experimental run.

7. Results of Experiments

7.1. RQ1: Universal Temporal Patterns Linking Physiology and Emotion

This section addresses the first research question: *Are there universal temporal patterns in which changes in physiological signals precede changes in emotional states (arousal, valence)?*

Table 3. ARMADA output statistics per dataset for both experimental runs.

Dataset	Participants	Patterns (Run 1)	Rules (Run 1)	Rules Raw (Run 2)	Rules Filtered (Run 2)
CASE	30	3469	2407	1617	371
K-emoCon	28	2948	2366	1666	432
CEAP	32	3028	2028	1776	465
EmoWorker_v2	31	2706	1851	1345	374

Cross-Dataset Rule Intersection

After applying semantic filters, 16 universal rules were identified as present in all four datasets. The top rules, sorted by average confidence, are presented in Table 4.

Table 4. Top universal rules found across all four datasets (sorted by average confidence).

Rule	Avg Conf	Avg Sup	CASE	K-emoCon	CEAP	EmoWorker
valence_low ⇒ before eda_peaks_high	0.721	0.632	0.30	0.86	0.97	0.77
valence_low ⇒ before eda_scr_auc_medium	0.718	0.628	0.74	0.81	0.66	0.67
hrv_cvnn_high ⇒ meets arousal_medium	0.708	0.568	0.80	0.68	0.58	0.77
hrv_cvsd_high ⇒ meets arousal_medium	0.708	0.564	0.76	0.62	0.65	0.81
hrv_pnn50_high ⇒ meets arousal_medium	0.702	0.530	0.86	0.54	0.62	0.79
hrv_rmssd_high ⇒ meets arousal_medium	0.691	0.529	0.70	0.56	0.75	0.76
hrv_sdnn_high ⇒ meets arousal_medium	0.682	0.509	0.80	0.58	0.58	0.76
valence_low ⇒ before arousal_medium	0.623	0.555	0.56	0.48	0.79	0.67

Key Findings

The results indicate that universal temporal patterns linking physiological signals to emotional states do exist across heterogeneous datasets. Two major categories of universal rules were identified:

HRV → Arousal rules. The most consistent universal pattern involves elevated HRV features (high CVNN, CVSD, PNN50, RMSSD, SDNN) temporally *meeting* medium arousal states. These rules appeared with average confidence between 0.68 and 0.71 across all four datasets, suggesting that periods of high parasympathetic cardiac activity reliably precede or co-occur with moderate arousal levels regardless of the experimental context.

Valence → EDA rules. Low valence states were found to temporally *precede* increases in electrodermal activity (high EDA peaks and medium SCR area under curve). These rules exhibited average confidence of 0.72, indicating that negative emotional valence acts as a precursor to heightened sympathetic arousal as reflected in skin conductance. The bidirectional nature of this pattern (emotion → physiology) is notable and will be further discussed in the context of the second research question.

8. Conclusion

8.1. Headings and Sections

Only the first two levels of section headings should be numbered. First-level headings need to be in 12-point bold type. Second-level headings should be in 11-point bold type. Third-level headings should be in 11-point bold italic. All headings should have 6-point paragraph spacing before and after.

The first paragraph immediately after a heading should not be indented. All subsequent paragraphs should be indented 0.6cm.

References to sections (as well as to figures, tables, etc.), should be capitalised, as in “As previously mentioned in Section 4.2, it was found that . . . ”.

8.2. Acknowledgements

An acknowledgments section may be added after the body text of the contribution. This section should not be numbered. Authors can employ this section to acknowledge assistance obtained from their colleagues and co-workers, financial support, permission to publish, etc. For the purposes of the initial submission, please do not provide any information in this section that might in any way compromise the independence of the anonymous peer-review process.

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If needed, authors may include appendices which follow the reference list. Each appendix should be lettered, for instance "Appendix A", "Appendix B", etc. If including long appendices, it may be better to include a permanent URL link to the document, e.g. a questionnaire instrument. Do not provide any URL in the initial submission that might provide information about the domain name of the authors' institutional affiliations.

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Please ensure that all illustrations are clear and legible. Figures and tables should be inserted in appropriate places throughout the body text. Please do not group them together at the beginning or at the bottom of the paper. Figures and tables need to be numbered sequentially. In contrast to the caption belonging to a figure, which should always appear under the illustration, the caption of a table should always be positioned above the illustration. Caption text should be in 9-point. Text in tables and figures must be clearly legible, no less than 8-point and no more than 9-point.

Table 5. Table captions should always be positioned above the illustration.

	Column 1	**Column 2**
Row A		
Row B		
Row C		

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Short captions should be centred between the margins while long captions, covering more than one line, should be justified. If captions do not constitute a full sentence, they should not have a period at the end. Leave whitespace of about 0.6 cm around the area covered by the figure/table and caption. Captions, labels, and other text in illustrations need to be in 9- point type.

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8.7. Equations and Program Code

Formulae are centred and numbered consecutively. The formulae numbers should be enclosed in parentheses to the right. An extra line should be placed above and below the displayed expression; for example:

$$x + y = z \tag{3}$$

Program code snippets should be set in typewriter font as follows:

```
function checkDateFormat($date)
{
  //match the format of the date
  if (preg_match ("/^([0-9]{4})-([0-9]{2})-([0-9]{2})$/", $date, $parts))
  {
    //check whether the date is valid or not
    if(checkDate($parts[2],$parts[3],$parts[1]))
      return true;
    else
      return false;
  }
  else
    return false;
}
```

8.8. Length of Contributions

Full papers must not exceed 12 pages, including references and appendices. Short papers must not exceed 8 pages, including references and appendices. Posters must not exceed 4 pages including references.

8.9. In-line Citations and Reference List

The reference section should appear at the end of the paper. References must be ordered alphabetically. Please make sure that all references are correctly listed. Only include references that are connected with the work presented in your paper. Write all references using the Latin alphabet.

[1] A footnote should appear like this. Please ensure that your footnotes are complete, fully punctuated sentences.

For citations in the text, use consecutive numbers placed in square brackets. The numbers in the reference section need to be without square brackets.

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